Quantitative Techniques – I
DMGT204

Long Ques:

1. Explain the Importance and Limitations of Statistics.

Ans: **Statistics**

"Statistics are classified facts respecting the conditions of the people in a state – especially those facts which can be stated in numbers or in tables of numbers or in any other tabular or classified arrangement."

The word 'statistics' has been used in both the plural and the singular sense. Statistics, in plural sense, means a set of numerical figures or data. In the singular sense, it represents a method of study

Importance of Statistics

The importance of statistics in the following major areas:

1. ***Importance to the State:*** In modern era, the role of State has increased and various governments of the world also take care of the welfare of its people. Therefore, these governments require much greater information in the form of numerical figures for the fulfillment of welfare objectives in addition to the efficient running of their administration.
2. ***Importance in economics:*** Statistics is an indispensable tool for a proper understanding of various economic problems. It also provides important guidelines for the formulation of various economic policies.
3. ***Importance in national income accounting:*** The system of keeping the accounts of income and expenditure of a country is known as national income accounting. These accounts contain information on various macro-economic variables like national income, expenditure, production, savings, investments, volume of exports and imports, etc. Since such information is very difficult, if not impossible, to obtain, is often estimated by using techniques and principles of statistics.
4. ***Importance in planning:*** Planning is indispensable for achieving faster rate of growth through the best use of a nation's resources. It also requires a good deal of statistical data on various aspects of the economy. The statistical analysis of data regarding an economy may reveal certain areas which might require special attention, e.g., a situation of growing unemployment or a situation of rising prices during past few years.
5. ***Importance in business:*** With the increase in size of business of a firm and with the uncertainties of business because of competition, the need for statistical information and statistical analysis of various business situations has increased tremendously.

Limitations of Statistics

The science of statistics suffers from the following limitations:

1. ***Statistics deals with numerical facts only:*** Quantitative facts are capable of being represented in the form of numerical figures and therefore, are also known as numerical facts. These facts can be analyzed and interpreted with the help of statistical methods.
2. ***Statistics deals only with groups and not with individuals:*** Statistical studies are undertaken to study the characteristics of a group rather than individuals. These studies are done to compare the general behaviour of the group at different points of time or the behaviour of different groups at a particular point of time. For example, the economic performance of a country in a year is measured by its national income in that year.

3. ***Statistical results are true only on the average:*** Statistical results give the behaviour of the group on the average and these may not hold for an individual of that very group. Thus, the statement that average wages of workers of a certain factory is 1,500 p.m. does not necessarily mean that each worker is getting this wage. In fact, some of the workers may be getting more while others less than or equal to 1,500.
4. ***Statistical results are only approximately true:*** Most of the statistical studies are based on a sample taken from the population. Under certain circumstances the estimated data are also used. Therefore, conclusions about a population based on such information are bound to be true only approximately.
5. ***Statistical methods constitute only one set of methods to study a problem:*** A given problem can often be studied in many ways. Statistical methods are used to simplify the mass of data and obtain quantitative results by its analysis. However, one should not depend entirely on statistical results.
6. ***Statistics are liable to be misused:*** Statistical data are likely to be misused to draw any type of conclusion. If the attitude of the investigator is biased towards a particular aspect of the problem, he is likely to collect only such data which give more importance to that aspect.
7. ***Statistics must be used only by experts:*** Statistics, being a technical subject, is very difficult for a common man to understand. Only the experts of statistics can use it correctly and derive right conclusions from the analysis.

2. Discuss the Classification and Presentation of Data? Explain the types of classification of Data.

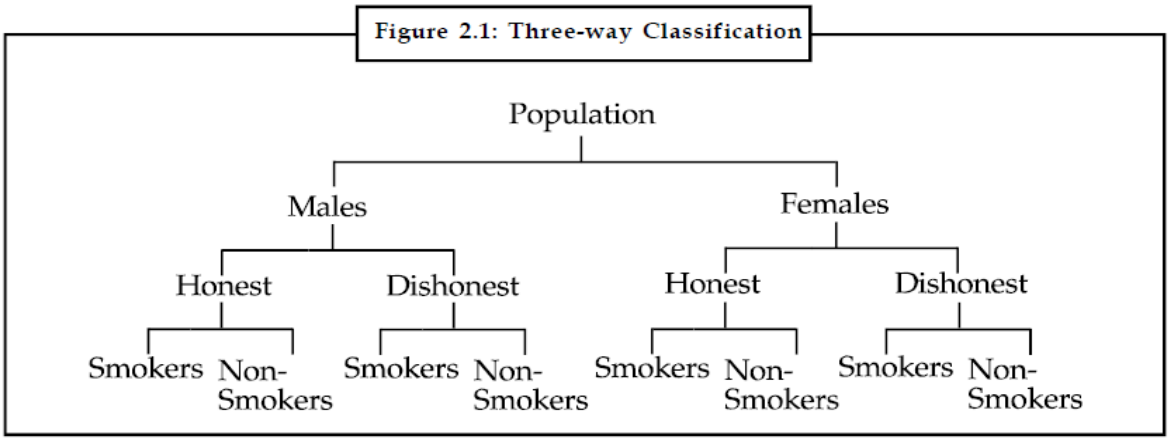
Ans: **Classification**

Classification is the process of arranging the available data into various homogeneous classes and subclasses according to some common characteristics or objective of investigation. In the words of L.R. Connor, "Classification is the process of arranging things (either actually or notionally) in the groups or classes according to the unity of attributes that may subsist amongst a diversity of individuals."

Types of Classification

The nature of classification depends upon the purpose and objective of investigation. The following are some very common types of classification:
1. ***Geographical (or spatial) classification:*** When the data are classified according to geographical location or region, it is called a geographical classification.
2. ***Chronological classification:*** When the data are classified on the basis of its time of occurrence, it is called a chronological classification. Various time series such as; National Income figures (annual), annual output of wheat, monthly expenditure of a household, daily consumption of milk, etc., are some examples of chronological classification.
3. ***Conditional classification:*** When the data are classified according to certain conditions, other than geographical or chronological, it is called a conditional classification.
4. ***Qualitative classification or classification according to attributes:*** When the characteristics of the data are non-measurable, it is called a qualitative data. The examples of non measurable characteristics are sex of a person, marital status, colour, honesty, intelligence, etc. If the data are classified only into two categories according to the presence or absence of an attribute, the classification is termed as dichotomous or twofold classification. On the other hand, if the data are classified into more than two categories according to an attribute, it is called a manifold classification. If the classification is done according to a single attribute, it is known as a one-way classification. On the other hand, the

classification done according to two or more attributes is known as a two-way or multiway classification respectively.



5. ***Quantitative classification or classification according to variables:*** In case of quantitative data, the characteristic is measurable in terms of numbers and is termed as variable, e.g., weight, height, income, the number of children in a family, the number of crime cases in a city, life of an electric bulb of a company, etc. A variable can take a different value corresponding to a different item of the population or universe. Variables can be of two types (a) Discrete and (b) Continuous.
(a) *Discrete Variable:* A discrete variable can assume only some specific values in a given interval. For example, the number of children in a family, the number of rooms on each floor of a multistoried building, etc.
(b) *Continuous Variable:* A continuous variable can assume any value in a given interval. For example, monthly income of a worker can take any value, say, between 1,000 to 2,500. The income of a worker can be 1,500.25, etc.

3. Discuss the methods of Collecting Data along with their merits and demerits.
Ans: **Collection of Data**
Data collection is the process of gathering and measuring information on variables of interest, in an established systematic fashion that enables one to answer stated research questions, test hypotheses, and evaluate outcomes. The data collection component of research is common to all fields of study including physical and social sciences, humanities, business, etc.

Method of Collecting Data
For collection of primary data, the investigator may choose any one or a combination of the following methods:

1. ***Direct Personal Observation:*** Under this method, the investigator collects data by having direct contact with the units of investigation. The accuracy of collected data depends, to a large extent, upon the training and attitude of the investigator and the supporting attitude of the respondents.
Merits

(a) Original data are collected.
(b) Collected data are more accurate and reliable.
(c) The collected data are often homogeneous and comparable.

Demerits

(a) This method is expensive and time consuming, particularly when the field of investigation is large.
(b) It is not possible to properly train a large team of investigators.
(c) Data are collected as per the convenience and willingness of the respondents.

2. ***Indirect Oral Interview:*** This method is used when the area of investigation is very large or the respondents are reluctant to part with the information due to various reasons. Under this method, the investigator collects data from a third party or witness or head of an institution, etc., who is supposed to be in touch with the respondents.

Merits

(a) It is economical in terms of time, money and manpower.
(b) Since the information is collected from the persons who are well aware of the situation, it is likely to be unbiased and reliable.
(c) This method is particularly suitable for the collection of confidential information. For example, a person may not like to reveal his habit of drinking, smoking, gambling, etc., which may be revealed by others.

Demerits

(a) As compared with direct personal observation, the degree of accuracy of the data is likely to be lower.
(b) The persons, providing the information, may be prejudiced or biased.
(c) The information collected from different persons may not be homogeneous and comparable.

3. ***Information Through Local Agencies or Correspondents:*** Under this method, local agents or correspondents are appointed in different parts of the area under investigation. These agents send the desired information at regular intervals of time. This method is often adopted by newspapers.

Merits

(a) This method is useful in situations where the area of investigation is very large and periodic information is to be collected from the distant places.
(b) It is economical in terms of time, money and labour.

Demerits

(a) The collected information lacks originality.
(b) The information is likely to be affected by the bias of the corespondents.
(c) It is not possible to obtain results with high degree of accuracy.

4. ***Information through Questionnaires Filled by Respondents:*** The information, in this method, is collected through the filling of questionnaires by the respondents. A questionnaire consists of a list of questions pertaining to the investigation. Blank spaces are left for writing answers. The questionnaire is sent to the respondents along with a covering letter for soliciting their cooperation by giving the correct information and returning the filled questionnaires well in time.

Merits

(a) This method is useful for the collection of information from an extensive area of investigation.

(b) This method is economical as it requires less time, money and labour.
(c) The collected information is original and more reliable.

Demerits

(a) The respondents may provide wrong information if the questions are not properly understood.
(b) It is not possible to collect information if the respondents are not educated.
(c) Since it is not possible to ask supplementary questions, the method is not flexible.

5. ***Information through Schedules Filled by Investigators:*** The work of filling of a questionnaire, termed as schedule here, can be done by the investigator through personal contact with the respondent. In order to get reliable information, the investigator should be tactful, well trained, unbiased and hard working.

Merits

(a) This method is suitable for an extensive area of investigation.
(b) Since the investigator has a direct contact with the respondents, it is possible to get accurate and reliable information.
(c) It is possible to get the information even if the respondents are not educated.

Demerits

(a) This method is very expensive and time consuming.
(b) The collected information is affected, to a large extent, by the bias of the investigator.
(c) If the investigators are negligent or not properly trained, the results of investigations are likely to be misleading

4. What do you mean by Drafting of Questionnaire? Explain in detail.

Ans: **Drafting a Questionnaire**

A questionnaire or a schedule is a list of questions relating to the problem under investigation. There is no basic difference between a questionnaire and a schedule. A questionnaire is filled by the respondent, while a schedule is filled by the investigator.
The quality of information collected through the filling of a questionnaire depends, to a large extent, upon the drafting of its questions. Hence, it is extremely important that the questions be designed or drafted very carefully and in a tactful manner.

The following general points must always be kept in mind to draft a good questionnaire.

1. The questions should be simple, unambiguous and precise.
2. The number of questions should be as small as possible. Only those questions which have a direct relevance to the problem be included.
3. The question should be framed in such a manner that their answers are specific and precise. The following four type of questions are generally framed in any questionnaire:
(a) *Simple alternative questions:* Such questions are answered by yes/no or right/wrong, etc.
(b) *Multiple choice questions:* In such questions, the possible answers are printed in the questionnaire and the respondent is supposed to tick any one of them.
(c) *Specific information questions:* Such questions are used to extract specific information like; how many members are there in your family, what is your monthly income, etc.
(d) *Open questions:* These types of questions are to be answered by the respondent in his own words. The questions should be such that it is possible to answer them in few words.
4. The questions should be capable of being easily answered by the respondents. The questions that rely too much on the memory of the respondent should be avoided.

5. The questions affecting the pride and sentiments of the respondents should be avoided. Similarly the questions pertaining to private affairs of the respondents should never be asked.
6. The questions relating to mathematical computations should be avoided.
7. The questions should follow a logical sequence so that a natural and spontaneous reply to each question is induced.
8. The questions should be directly related to the objective(s) of investigation.
9. Certain corroborative questions must also be asked to verify the accuracy of the supplied information.
10. Necessary instructions for filling the questionnaire should also be given in simple and precise form.
11. Enough space should be provided for answers. The questionnaire should look as attractive as possible.
12. Before actually using a questionnaire, a test check must always be done by obtaining answers from some respondents. The questionnaire should be modified, if necessary, in the light of these answers.

5. Discuss the measures of central tendency along with merits and demerits.

Ans: **Central Tendency**

Summarization of the data is a necessary function of any statistical analysis. As a first step in this direction, the huge mass of unwieldy data is summarized in the form of tables and frequency distributions. A measure of central tendency or an average is very essential and an important summary measure in any statistical analysis.

Measures of Central Tendency

- **Arithmetic Mean**

It will be assumed that there are n observations whose values are denoted by X1,X2, ... Xn respectively. The sum of these observations X1 + X2 + ... + Xn will be denoted in abbreviated form as, where $\sum$ (called sigma) denotes summation sign.
Arithmetic Mean is defined as the sum of observations divided by the number of observations. It can be computed in two ways:
(i) Simple arithmetic mean
(ii) Weighted arithmetic mean. In case of simple arithmetic mean, equal importance is given to all the observations while in weighted arithmetic mean, the importance given to various observations is not same.

Merits

1. Arithmetic mean is rigidly defined by an algebraic formula.
2. Calculation of arithmetic mean is based on all the observations and hence, it can be regarded as representative of the given data.
3. It is capable of being treated mathematically and hence, is widely used in statistical analysis.
4. Arithmetic mean can be computed even if the detailed distribution is not known but sum of observations and numbers of observations are known.
5. It is least affected by the fluctuations of sampling.

Demerits

1. It can neither be determined by inspection nor by graphical location.
2. Arithmetic mean cannot be computed for a qualitative data; like data on intelligence, honesty, smoking habit, etc.
3. It is too much affected by extreme observations and hence, it does not adequately represent data consisting of some extreme observations.

4. In the absence of a complete distribution of observations the arithmetic mean may lead to confusions. For example, there may be two entirely different distributions with same value of arithmetic mean.
5. Simple arithmetic mean gives greater importance to larger values and lesser importance to smaller values.

- **Median**

Median of distribution is that value of the variate which divides it into two equal parts. In terms of frequency curve, the ordinate drawn at median divides the area under the curve into two equal parts. Median is a positional average because its value depends upon the position of an item and not on its magnitude.

Merits

1. It is easy to understand and easy to calculate, especially in series of individual observations and ungrouped frequency distributions. In such cases it can even be located by inspection.
2. Median can be determined even when class intervals have open ends or not of equal width.
3. It is not much affected by extreme observations. It is also independent of range or dispersion of the data.
4. Median can also be located graphically.
5. Median conveys the idea of a typical observation.

Demerits

1. It, being a positional average, is not capable of being treated algebraically.
2. In case of individual observations, when the number of observations is even, the median is estimated by taking mean of the two middle-most observations, which is not an actual observation of the given data.
3. It is not based on the magnitudes of all the observations. There may be a situation where different sets of observations give same value of median.
4. In comparison to arithmetic mean, it is much affected by the fluctuations of sampling.
5. Since it is not possible to define weighted median like weighted arithmetic mean, this average is not suitable when different items are of unequal importance.

- **Mode**

Mode is that value of the variate which occurs maximum number of times in a distribution and around which other items are densely distributed. The concept of mode, as a measure of central tendency, is preferable to mean and median when it is desired to know the most typical value, e.g., the most common size of shoes, the most common size of a ready-made garment, the most common size of income, etc. "Mode is the value which has the greatest frequency density in the given frequency data."

Merits

1. It is easy to understand and easy to calculate. In many cases it can be located just by inspection.
2. Like mean or median, it is not affected by extreme observations. It can be calculated even if these extreme observations are not known.
3. It can be determined even if the distribution has open end classes.
4. It can be located even when the class intervals are of unequal width provided that the width of modal and that of its preceding and following classes are equal.
5. It is a value around which there is more concentration of observations and hence the best representative of the data.

Demerits

1. It is not based on all the observations.
2. It is not capable of further mathematical treatment.

3. In certain cases mode is not rigidly defined and hence, the important requisite of a good measure of central tendency is not satisfied.
4. It is much affected by the fluctuations of sampling.
5. It is not suitable when different items of the data are of unequal importance.

6. What is correlation? Discuss the various types of Correlation with the help of scatter diagram.

Ans: **Correlation**

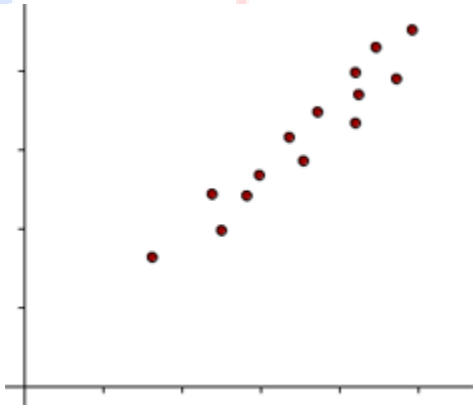
"Correlation is an analysis of co variation between two or more variables."

In statistics, the degree of association between two RANDOM VARIABLES. The correlation between the graphs of two data sets is the degree to which they resemble each other.

Types of Correlation

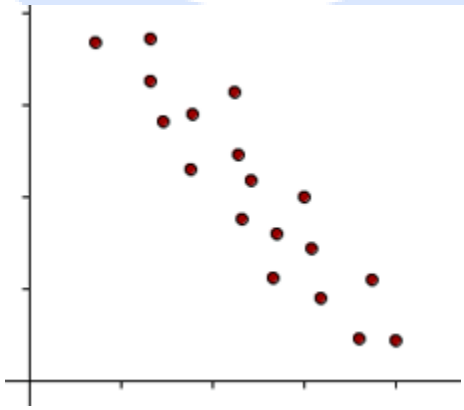
Positive Correlation

Positive correlations occur when an increase in one variable increases the value in another. The line corresponding to the scatter plot is an increasing line.



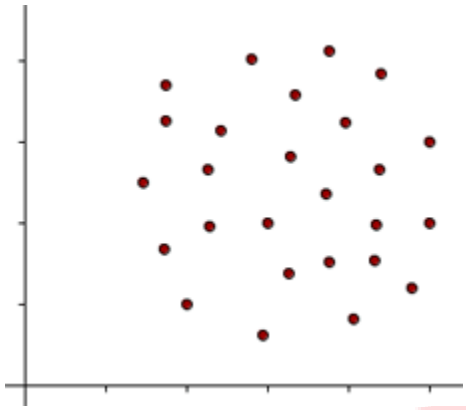
Negative Correlation

Negative correlation occurs when an increase in one variable decreases the value of another. The line corresponding to the scatter plot is a decreasing line.



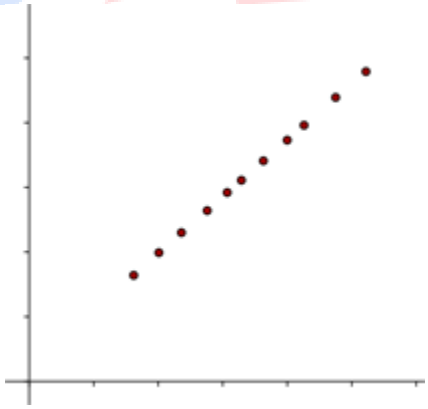
No Correlation

No correlation occurs when there is no linear dependency between the variables.



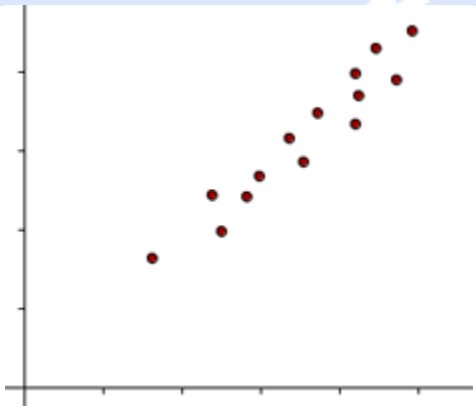
Perfect Correlation

Perfect correlation occurs when there is a functional dependency between the variables. In this case all the points are in a straight line.



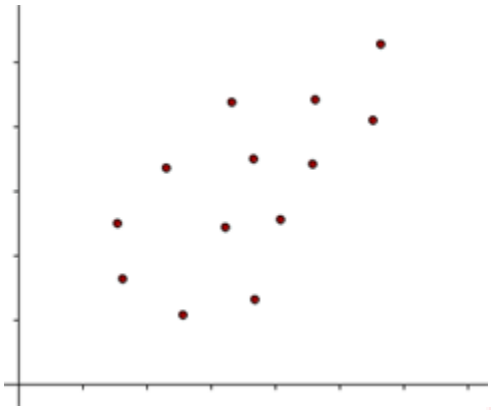
Strong Correlation

A correlation is stronger the closer the points are located to one another on the line.



Weak Correlation

A correlation is weaker the farther apart are located to one another on the line.



7. Define Index number with their characteristics and discuss the various methods of construction of Index numbers.

Ans: **Index Numbers**

Index numbers are meant to study the change in the effects of such factors which cannot be measured directly. According to Bowley, "Index numbers are used to measure the changes in some quantity which we cannot observe directly". For example, changes in business activity in a country are not capable of direct measurement but it is possible to study relative changes in business activity by studying the variations in the values of some such factors which affect business activity, and which are capable of direct measurement.

Definition and Characteristics of Index Numbers

"An index number is a special type of average that provides a measurement of relative changes from time to time or from place to place." OR

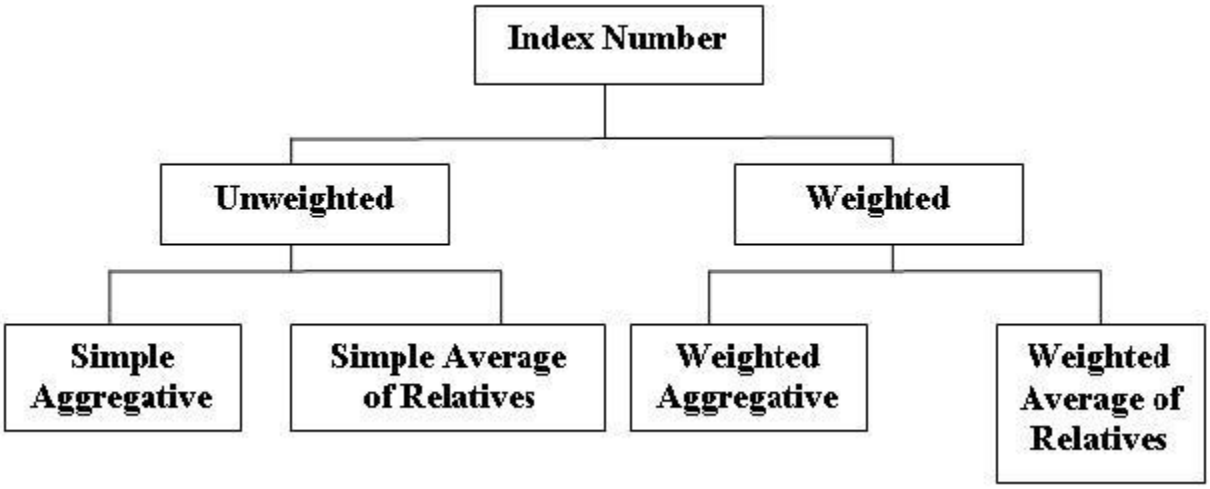
"An index number is a device for comparing the general level of magnitude of a group of distinct, but related, variables in two or more situations."

On the basis of the above definitions, the following characteristics of index numbers are:

1. ***Index numbers are specialized averages:*** As we know that an average of data is its representative summary figure. In a similar way, an index number is also an average, often a weighted average, computed for a group. It is called a specialized average because the figures, that are averaged, are not necessarily expressed in homogeneous units.
2. ***Index numbers measure the changes for a group which are not capable of being directly measured: The examples of such magnitudes are:*** Price level of a group of items, level of business activity in a market, level of industrial or agricultural output in an economy, etc.
3. ***Index numbers are expressed in terms of percentages:*** The changes in magnitude of a group are expressed in terms of percentages which are independent of the units of measurement. This facilitates the comparison of two or more index numbers in different situations.

Methods of Index Number Construction

The methods of construction of index number are classified in the following table:-



8. What is the meaning and objectives of Time Series Analysis? Discuss the methods of Time Series Analysis.

Ans: **Analysis of Time Series**

A time series is a collection of observations of well-defined data items obtained through repeated measurements over time. For example, measuring the value of retail sales each month of the year would comprise a time series. This is because sales revenue is well defined, and consistently measured at equally spaced intervals. Data collected irregularly or only once are not time series. A series of observations, on a variable, recorded after successive intervals of time is called a time series.

Objectives of Time Series Analysis

The analysis of time series implies its decomposition into various factors that affect the value of its variable in a given period. It is a quantitative and objective evaluation of the effects of various factors on the activity under consideration. There are two main objectives of the analysis of any time series data:

1. To study the past behaviour of data.
2. To make forecasts for future.

The study of past behaviour is essential because it provides us the knowledge of the effects of various forces. This can facilitate the process of anticipation of future course of events and, thus, forecasting the value of the variable as well as planning for future.

Methods of Time Series Analysis

1. ***Method of Selected Points:*** In this method, two points, considered to be the most representative or normal, are joined by a straight line to get secular trend. This, again, is a subjective method since different persons may have different opinions regarding the representative points. Further, only linear trend can be determined by this method.
2. ***Method of Semi-averages:*** The given time series data are divided into two equal parts and the arithmetic mean of the values of each part is computed. The computed means are termed as semi-averages. Each semi-average is paired with the centre of time period of its part. The two pairs are then plotted on a graph paper and the points are joined by a straight line to get the trend. It should be pointed out here that in case of odd number of observations, the two equal parts are obtained by dropping the middle most observation.

Merits

(a) It is simple method of measuring trend.

(b) It is an objective method because anyone applying this to a given data would get identical trend values.

Demerits

(a) This method can give only a linear trend of the data irrespective of whether it exists or not.
(b) This is only a crude method of measuring trend, since we do not know whether the effects of other components is completely eliminated or not.

3. ***Method of Moving Average:*** In the method of moving average, successive arithmetic averages are computed from overlapping groups of successive values of a time series. Each group includes all the observations in a given time interval, termed as the period of moving average. The next group is obtained by replacing the oldest value by the next value in the series. The averages of such groups are known as the moving averages.

Merits

(a) This method is easy to understand and easy to use because there are no mathematical complexities involved.
(b) It is an objective method.
(c) It is a flexible method in the sense that if a few more observations are added, the entire calculations are not changed.

Demerits

(a) It is not possible to calculate trend values for all the items of the series. Some information is always lost at its ends.
(b)The selection of period of moving average is a difficult task and a great deal of care is needed to determine it.
(c) Like arithmetic mean, the moving averages are too much affected by extreme values.

9. Define Probability and discuss the Probability Theorems.

Ans:" **Probability** is the chance that something will happen - how likely it is that some event will happen."

If n is the number of equally likely, mutually exclusive and exhaustive outcomes of a random experiment out of which m outcomes are favourable to the occurrence of an event A, then the probability that A occurs, denoted by P(A), is given by:

$$\text{Probability}(A) = \frac{\text{Total number of favourable outcomes of A}}{\text{No. of total outcomes}}$$

Probability Theorems

Theorem 1: Expected value of a constant is the constant itself, i.e., E(b) = b, where b is a constant.

Theorem 2: E(aX) = aE(X), where X is a random variable and a is constant.

Expected Monetary Value (EMV)

When a random variable is expressed in monetary units, its expected value is often termed as expected monetary value and symbolized by EMV. Expected Monetary Value (EMV) is the weighted average of probable outcomes. It represents the expected average payoff if you made that decision using the same payoffs and probabilities an infinite number of times.

Expectation of the Sum or Product of two Random Variables

Theorem 1: If X and Y are two random variables, then E(X + Y) = E(X) + E(Y).

Theorem 2: If X and Y are two independent random variables, then E(X.Y) = E(X).E(Y)

Expectation of a Function of Random Variables

To gain further insights about the behavior of random variables, we first consider their expectation, which is also called mean value or expected value. The definition of expectation follows our intuition.

Theorem 1 (Expectation) Let X and Y be random variables with finite expectations.

1. If g(x) ≥ h(x) for all x ∈ R, then E[g(X)] ≥ E[h(X)].
2. E(aX + bY + c) = aE(X) + bE(Y) + c for any a, b, c ∈ R.

10. Discuss the measures of Dispersion.

Ans: **Measures of Dispersion**

The measures of central tendency are not adequate to describe data. Two data sets can have the same mean but they can be entirely different. Thus to describe data, one needs to know the extent of variability. This is given by the measures of dispersion. Range, interquartile range, and standard deviation are the three commonly used measures of dispersion.

The following are some important measures of dispersion

(a) Range

The range is the difference between the largest and the smallest observation in the data. The prime advantage of this measure of dispersion is that it is easy to calculate. On the other hand, it has lot of disadvantages. It is very sensitive to outliers and does not use all the observations in a data set. It is more informative to provide the minimum and the maximum values rather than providing the range.

Merits

1. It is easy to understand and easy to calculate.
2. It gives a quick measure of variability.

Demerits

1. It is not based on all the observations.
2. It is very much affected by extreme observations.
3. It only gives rough idea of spread of observations.

(b) Inter-Quartile Range

Interquartile range is defined as the difference between the 25th and 75th percentile (also called the first and third quartile). Hence the interquartile range describes the middle 50% of observations. If the interquartile range is large it means that the middle 50% of observations are spaced wide apart. The important advantage of interquartile range is that it can be used as a measure of variability if the extreme values are not being recorded exactly (as in case of open-ended class intervals in the frequency distribution). Other advantageous feature is that it is not affected by extreme values. The main disadvantage in using interquartile range as a measure of dispersion is that it is not amenable to mathematical manipulation.

Merits

1. It is rigidly defined.
2. It is easy to understand and easy to compute.
3. It is not affected by extreme observations and hence a suitable measure of dispersion when a distribution is highly skewed.
4. It can be calculated even for a distribution with open ends.

Demerits

1. Since it is not based on all the observations, hence, not a reliable measure of dispersion.
2. It is very much affected by the fluctuations of sampling.
3. It is not capable of being treated mathematically.

(c) Mean Deviation

Mean deviation is the average differences among the items in a series from the mean itself or median or mode of that series. It is concerned with the extent of which the values are dispersed about the mean or median or the mode. It is found by averaging all the deviations from control tendency. These deviations are taken into computations with regard to negative sign. Theoretically the deviations of item are taken preferably from median instead than from

the mean and mode.

Merits

1. It is easy to understand and easy to compute.
2. It is based on all the observations.
3. It is less affected by extreme observations vis-a-vis range or standard deviation (to be discussed in the next section).
4. It is not much affected by fluctuations of sampling.

Demerits

1. It is not capable of further mathematical treatment. Since mean deviation is the arithmetic mean of absolute values of deviations, it is not very convenient to be algebraically manipulated.
2. This necessitates a search for a measure of dispersion which is capable of being subjected to further mathematical treatment.
3. It is not well defined measure of dispersion since deviations can be taken from any measure of central tendency.

(d) Standard Deviation

Standard deviation (SD) is the most commonly used measure of dispersion. It is a measure of spread of data about the mean. SD is the square root of sum of squared deviation from the mean divided by the number of observations.

Merits

1. It is a rigidly defined measure of dispersion.
2. It is based on all the observations.
3. It is capable of being treated mathematically. For example, if standard deviations of a number of groups are known, their combined standard deviation can be computed.
4. It is not very much affected by the fluctuations of sampling and, therefore, is widely used in sampling theory and test of significance.

Demerits

1. As compared to the quartile deviation and range, etc., it is difficult to understand and difficult to calculate.
2. It gives more importance to extreme observations.
3. Since it depends upon the units of measurement of the observations, it cannot be used to compare the dispersions of the distributions expressed in different units.

Short Ques:

1. What do you mean by Statistics?

Ans: The practice or science of collecting and analyzing numerical data in large quantities, especially for the purpose of inferring proportions in a whole from those in a representative sample.

'OR'The science that deals with the collection, classification, analysis, andinterpretation of nu merical facts or data, and that, by use of mathematical theories of probability, imposes order and regularity on aggregates of more or less disparate elements.

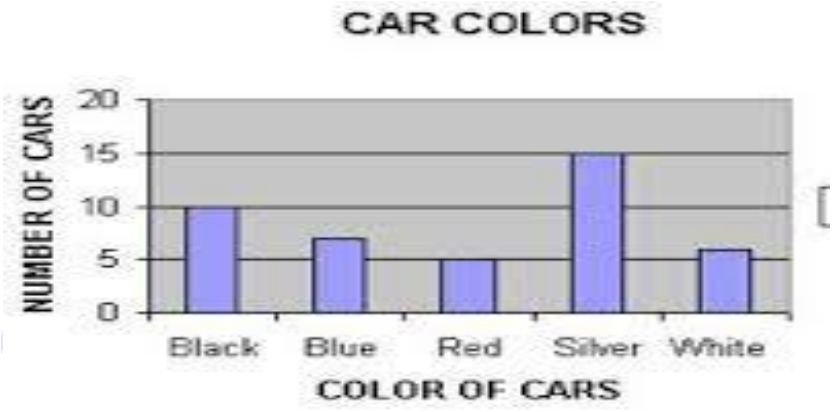
2. Explain the Classification of Tables.

Ans: The collected data are a complex and unorganized mass of figures which is very difficult to analyze and interpret. Therefore, it becomes necessary to organize this so that it becomes easier to grasp its broad features. Further, in order to apply the tools of analysis and

interpretation, it is essential that the data are arranged in a definite form. This task is accomplished by the process of classification and tabulation.

3. Define Bar Diagram and also draw its diagram.

Ans: A bar chart or bar graph is a chart with rectangular bars with lengths proportional to the values that they represent. The bars can be plotted vertically or horizontally. A vertical bar chart is sometimes called a column bar chart.



4. What do you mean by Regression Analysis

Ans: **Regression analysis** is a statistical process for estimating the relationships among variables. It includes many techniques for modeling and analyzing several variables, when the focus is on the relationship between a dependent variable and one or more independent variables.

5. Give difference between Primary and Secondary Data?

Ans: **Primary Data**

1. Primary data are always original as it is collected by the investigator.
2. Suitability of the primary data will be positive because it has been systematically collected.
3. Primary data are expensive and time consuming.
4. Extra precautions are not required.
5. Primary data are in the shape of raw material.
6. Possibility of personal prejudice.

Secondary Data

1. Secondary data lacks originality. The investigator makes use of the data collected by other agencies.
2. Secondary data may or may not suit the objects of enquiry.
3. Secondary data are relatively cheaper.
4. It is used with great care and caution.
5. Secondary data are usually in the shape of readymade products.
6. Possibility of lesser degree of personal prejudice.

6. What is Expected monetary value?

Ans: Total of the weighted outcomes (payoffs) associated with a decision, the weights reflecting the probabilities of the alternative events that produce the possible payoff. It is expressed mathematically as the product of an event's probability of occurrence and the gain or loss that will result.

7. What do you mean by Permutation and Combination?

Ans: **Permutation**

The arrangement made by taking some or all elements out of a number of things is called a permutation.

Combination

The group or selection made by taking some or all elements out of a number of things is called a combination.

8. Write the formulae for (i) Karl Pearson's Coefficient of Correlation and (ii) Spearman's Rank Correlation.

Ans: **Karl Pearson's Coefficient of Correlation**

$$r = \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{\sqrt{[\, n\Sigma x^2 - (\Sigma x)^2 \,][\, n\Sigma y^2 - (\Sigma y)^2 \,]}}$$

Spearman's Rank Correlation

$$r = 1 - \frac{6\sum d^2}{n^3 - n}$$

9. Explain the fitting to linear trend according to least square method.

Ans: **Least Square Methods**

This is one of the most popular methods of fitting a mathematical trend. The fitted trend is termed as the best in the sense that the sum of squares of deviations of observations, from it, are minimised.

Fitting of Linear Trend

Given the data (Yt, t) for n periods, where t denotes time period such as year, month, day, etc., we have to find the values of the two constants, a and b, of the linear trend equation Yt = a + bt.

10. "All periodic variations are not necessary seasonal." Comment.

Ans: **Periodic Variations*:*** These variations, also known as oscillatory movements, repeat themselves after a regular interval of time. This time interval is known as the period of oscillation. The oscillatory movements are termed as Seasonal Variations if their period of oscillation is equal to one year and as Cyclical Variations if the period is greater than one year. Although the periodic variations are more or less regular, they may not necessarily be uniformly periodic, i.e., the pattern of their variations in different periods may or may not be identical in respect of time period and size of periodic variations.